

Compare and contrast Mitosis and Meiosis Worksheet

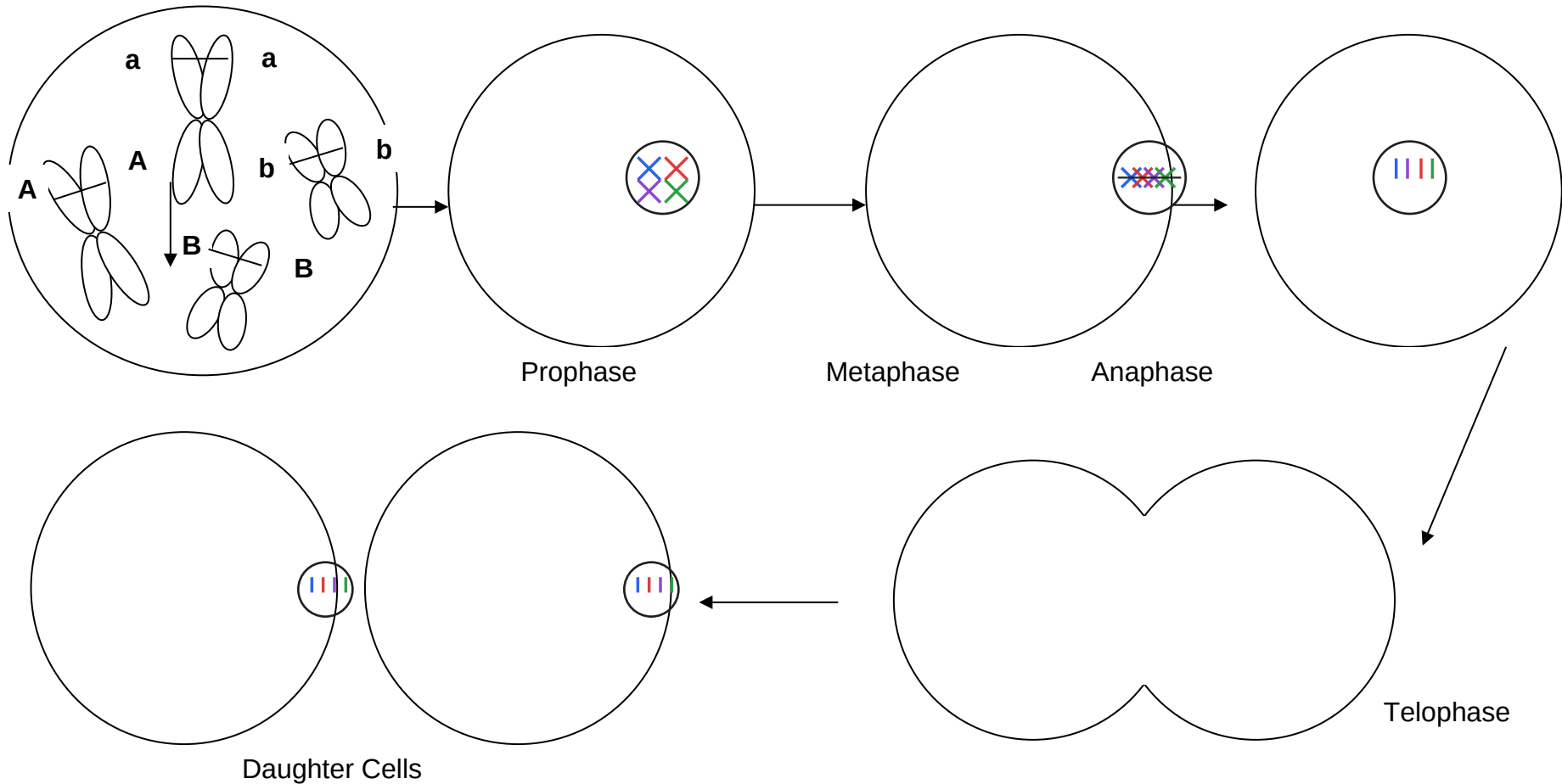
Part I Mitosis → Use Diploid skin cells for Mitosis with $2n = 4$ (as shown in picture below)

Material:

colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

- I. Draw the chromosomes in the cell as it undergoes **Mitosis**:
Remember homologous chromosomes line up on separate spindle fiber



Part II Meiosis

II. Draw the chromosomes in the cell as it undergoes Meiosis (same parent cell):

Use Use Diploid testical cells for Mitosis with $2n = 4$ (as shown in the picture below)

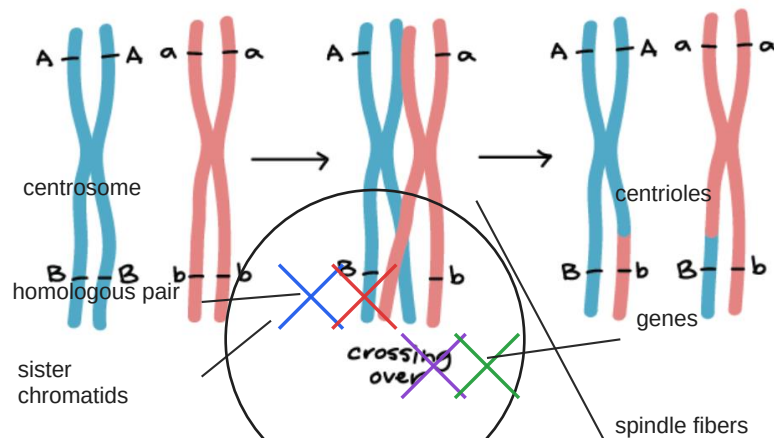
Remember Homologous chromosomes line up on separate spindle fibers

Material:

colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

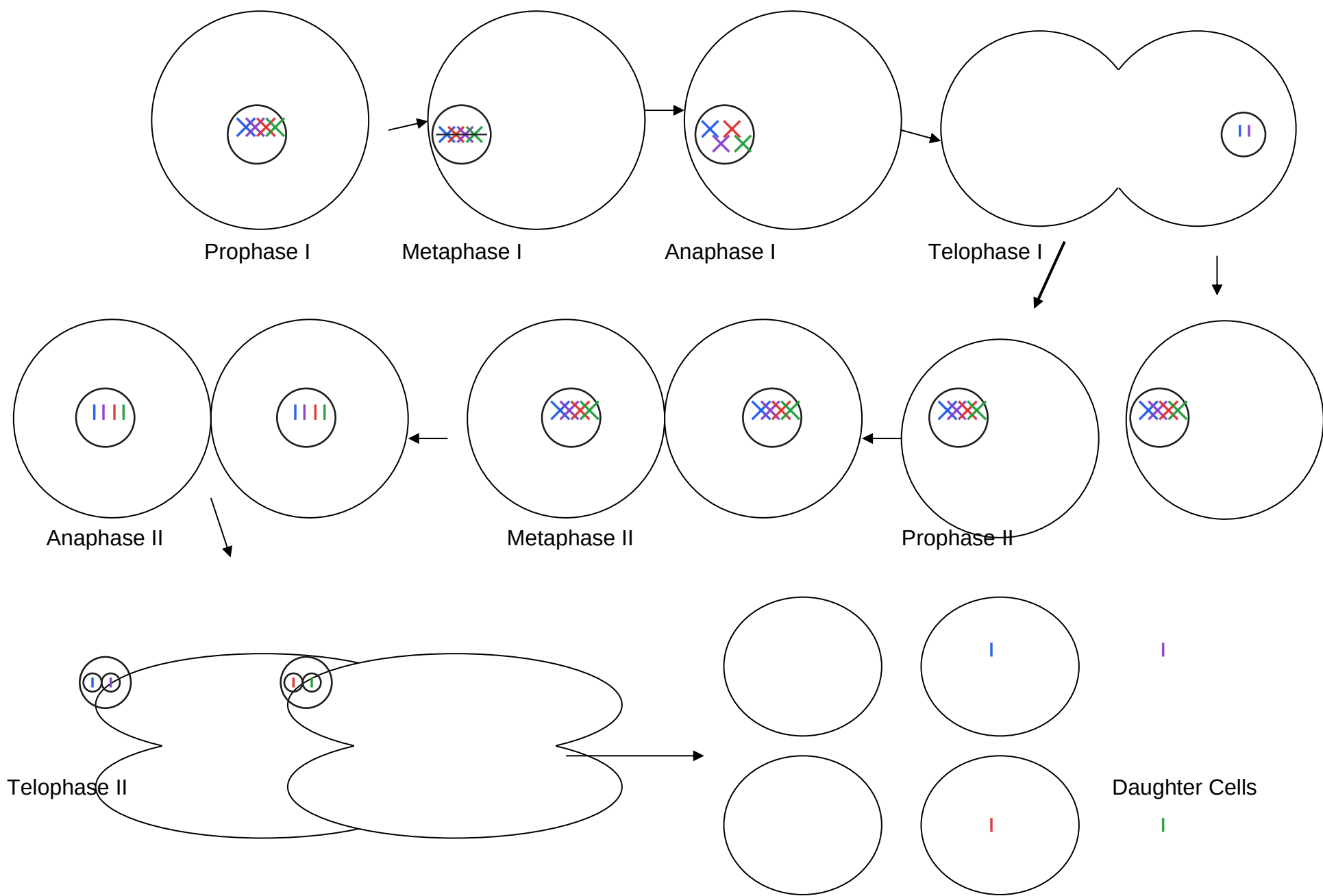
Show crossing over by switching the genes during Prophase I of meiosis



Note that chromosomes A, a are on one homologous pair of chromosomes in our exercise, and B, b are on a separate second homologous pair of chromosomes

Label the following structures: **homologous chromosomes, sister chromatids, genes, spindle fibers, centrosome, centrioles, and the nuclear membrane.**

You do **not** need to include all labels on a single cell—spread them across the diagram so each structure is clearly visible and easy to identify.



Part III: Complete the summary chart Comparing and Contrasting Mitosis and Meiosis

Cell Processes	Mitosis	Meiosis
Creates	body cells	sex cells (gametes)
Definition	1 division -> 2 identical 2n cells	2 divisions -> 4 different n cells
End Products	2 diploid cells	4 haploid cells
Steps	PMAT once	PMAT I + PMAT II
Type of Reproduction	asexual	sexual

Are they identical to the parent cell?	yes	no
When does cytokinesis occur?	once (after telophase)	twice (after I and II)
How many times does the parent cell divide?	1 time	2 times
What happens to the number of chromosomes at the end of each process? Are they in pairs or individual chromosomes?	stays $2n$ (46 in humans) in pairs	halves to n (23 in humans) individual
Why is each important?	growth + repair	variation + proper # for fertilization

How many chromosomes do human body cells and human sex cells have after they go through each process?	Mitosis: body cells = 46 Meiosis: sex cells = 23	Mitosis keeps chromosome number the same ($2n$). Meiosis halves it to n for gametes.
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Part III

Explain **nondisjunction**—when chromosomes or chromatids fail to separate properly—and how it can lead to conditions such as **Down syndrome** and **Turner syndrome**.

Then explain how these chromosomal abnormalities result in different physical and developmental effects.

Nondisjunction = chromosomes do not separate right.

In meiosis I homologous chromosomes can fail to split;

meiosis II sister chromatids can fail to split.

This makes gametes with extra or missing chromosomes.

Down syndrome: trisomy 21 (47 total).

Turner syndrome: monosomy X (45, X).

These change gene dosage, so development can differ:

Down can include learning delays and heart risks;

Turner can include short stature and ovarian issues.